Lab M11.03 – Pipeline Type Decision Lab (Problem Lab)

Data Flow 1: Transaction Fraud Detection

Pipeline Decision

- Pipeline Type: Streaming
- Architecture: ETL
- Latency Target: Sub-second
- Target System: Data Warehouse

Justification

1. Why this pipeline type?

   Transactions must be controlled continuously to detect fraud

2. Why this architecture (ETL/ELT)?

   Compliance or regulatory rules apply. Strong validation is required before storage. The "Transformation" step (Tokenization) must happen before the data reaches any database

3. Data flow description:

   - Source → Extract → Transform (tokenization), ML Inference Engine → Load → Data Warehouse

Failure Strategy

- What fails? Tokenizer service failure
- Recovery plan:
- Idempotency: can run multiple times without duplicating because of the transaction_id

Trade-offs

- Chose Streaming over Batch: Batch would be useless for preventing fraud in real-time.
- Risk of this approach: High operational complexity of streaming infrastructure.
- Mitigation: Use managed cloud services.

Data Flow 2: Daily Financial Reporting

Pipeline Decision

- Pipeline Type: Batch

- Architecture: ETL

- Latency Target: Daily

- Target System: Data Warehouse

Justification

1. Why this pipeline type?

Financial reporting is aggregate-heavy and requires the data just once at the end of the day.

2. Why this architecture (ETL/ELT)?

SOX compliance. Transforming data in a staging environment before loading it into the warehouse ensures only "clean" data is used for financial reporting.

3. Data flow description:

- Source → Extract → Transform (Validation/Aggregation) → Load → Data Warehouse

Failure Strategy

- What fails? Database connection, transformation logic error

- Recovery plan: Re-run the daily batch job.

- Idempotency: Yes, overwrite the daily analysis.

Trade-offs

- Chose ETL over ELT: ELT risks loading raw or un-audited data into the warehouse, which is a major compliance risk for financial reporting

- Risk of this approach: High dependency on the transformation logic accuracy

- Mitigation: checks, comparing source record counts with loaded record counts

Data Flow 3: Customer 360 Profile

Pipeline Decision

- Pipeline Type: Batch
- Architecture: ELT
- Latency Target: Weekly
- Data lake

Justification

1. Why this pipeline type?

Marketing needs weekly data

2. Why this architecture (ETL/ELT)?

We need to capture raw data first to handle PII compliance/GDPR consent tracking. Transformations happen ilater to allow flexibility in how we build profiles.

3. Data flow description:

- Source → Extract → Load (bronze, raw) → Transforms (silver, gold) → Datalake (gold) dashboards

Failure Strategy

- What fails? API fails
- Recovery plan: re-ingest raw data from Bronze layer if transformation fails
- Idempotency: Upsert logic

Trade-offs

- Chose ELT over ETL: variety of data makes early transformation difficult. ELT is more scalable.
- Risk of this approach:
- Mitigation:

Data Flow 4: Application Logs

Pipeline Decision

- Pipeline Type: Streaming
- Architecture: ELT
- Latency Target: Sub-second / minutes
- Target System: Data lake

Justification

1. Why this pipeline type?

SREs need alerts immediately to fix problems

2. Why this architecture (ETL/ELT)?

Logs are high-volume and variable. Extracting and loading raw JSON logs allows SREs to parse them dynamically as needed for debugging.

3. Data flow description:

- Source → Extract → Load → Transform (debugging) → Target

Failure Strategy

- What fails?
- Recovery plan:
- Idempotency: logs are appended

Trade-offs

- Chose ELT over ETL: Transforming 10M events/day before ingestion would create a massive bottleneck
- Risk of this approach: High storage costs for raw logs
- Mitigation: Implement retention policies

Data Flow 5: Partner Data Ingestion

Pipeline Decision

- Pipeline Type: Batch
- Architecture: ELT
- Latency Target: Weekly
- Target System: Data Warehouse

Justification

1. Why this pipeline type?

   - Source is an SFTP file delivered weekly. There is no need for anything faster.

2. Why this architecture (ETL/ELT)?

   - Simple structure. Load raw files to a staging area, then transform in the warehouse

3. Data flow description:

   - Source → Extract → Load → Transform → Bi reports

Failure Strategy

- What fails? SFTP file missing, incorrect file format
- Recovery plan: Automated monitoring for file arrival. If it fails, send notification to the Data Engineer for manual intervention.
- Idempotency: replacing

Trade-offs

- Chose ELT over ETL: more flexible and reprocessing is easier
- Risk of this approach: raw data stored, partner NDA non-compliant
- Mitigation: Set up automated file-arrival alerts

Data Flow	Pipeline Type	Architecture	Latency Target	Target	Key Risk
1. Fraud Detection	Streaming	ETL	Sub-second	Real-time ML	High operational complexity
2. Financial Reporting	Batch	ETL	Daily	Data Warehouse (Gold Layer)	High dependency on the transformation
3. Customer 360	Batch	ELT	Weekly	Data Lake (Silver/Gold)	PII/GDPR compliance
4. Application Logs	Streaming	ELT	Minutes	Data Lake (Silver)	High storage costs
5. Partner Data	Batch	ELT	Weekly	BI, data warehouse (gold)	Partner dependency